

$$1. \mu (\text{masses of cucumbers}) = 310 \text{ gm}$$

$$\sigma (\text{standard deviation}) = 22 \text{ gm}$$

$$n (\text{New cucumbers}) = 40$$

$$\mu (\text{new}) = 316$$

$$\alpha = 5\%$$

To test the claim, we will count the Z_i Test

$$Z_i = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}, \text{ where } \bar{x} = \mu (\text{new}) = 316$$

$$\mu = \mu = 310$$

$$\sigma = 22$$

$$\sqrt{n} = \sqrt{40}$$

$$Z_i = \frac{316 - 310}{\frac{22}{\sqrt{40}}} = \frac{6}{22\sqrt{40}} = \frac{6}{3.478} = 1.728$$

Define the hypothesis

$$N_{H_0} \text{ hypothesis } (H_0) \mu = 310 \text{ gm}$$

$$A_{H_1} \text{ hyp } (H_1) \mu > 310 \text{ gm}$$

Let Hyp: The new plant food has increased the mass. $\therefore$ we will find the Z_i critical value @ 0.95
i.e. 1.645

Since 1.728 is > 1.645

$\therefore$ we can conclude $Z_i \text{ score} > Z_i \text{ critical}$

so, we reject the Null Hypothesis
Statistical evidence support that new plant food claim made by its producers.

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Assignment 2

Section 2

2. To test if the new course was useful

No. of students = 12

Marks obtained before & after the test

Before : 44 40 61 52 32 44 70 41 67 72 53 72

After : 53 38 69 57 46 39 73 48 73 74 60 78

Diff: 9 -2 8 5 14 -5 3 7 6 2 7 6

H_0 : New course has no impact on marks

H_1 : New course has significant

Assume : $\alpha = 0.25$

Mean of differences : $(\bar{x}_d) = \frac{50}{12} = 4.17$

S.D. of differences : $(s_d) = 4.08$

Standard error : $(s_e) = \frac{s_d}{\sqrt{n}} = \frac{4.08}{\sqrt{12}}$

Calculate T-stat = $\frac{\bar{x}_d}{\frac{s_d}{\sqrt{n}}} = \frac{\bar{x}_d}{\text{Standard error}} = \frac{4.17}{\frac{4.08}{3.464}} = 3.54$

degree of freedom = $Sf = 11$

T-critical @ α as 0.95 for $df = 11$ is 1.796

$T_{\text{stat}} > T_{\text{critical}}$

We reject Null hypothesis
course has significant impact on marks.

Assumption
we are
considering
on
five significant
impact
hence
only tail
test

3.

Shifts	Good	Bad	Total
Day	900	130	1030
Evening	700	170	870
Night	400	200	600
Total	2000	500	2500

Test if there is any association between shifts & quality

We will apply Chi-square Test

$$\text{Expected Value} = \frac{\text{Row Total} \times \text{Col. Total}}{\text{Grand Total}}$$

$\therefore$ Expected value would be

$$O(D, G) = \frac{1030 \times 2000}{2500} = 824$$

$\therefore$ Expected Table would be

Shifts	Good	Bad
Day	824	206
Evening	696	174
Nights	480	120

$$\chi^2_{\text{square}} = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$= \sum_{\text{Day}} + \sum_{\text{Evening}} + \sum_{\text{Nights}}$$

$$= 101.83$$

$$DF = (R-1)(C-1) = 2$$

Critical χ^2 value for $\alpha @ 0.95$ & $DF = 2 = 5.991$

$\therefore$ we reject the Null Hypothesis
Hence we can conclude there is a significant association b/w shifts & quality.

Section 2

4. One-way - Anova

Hank	Joseph	Susan
8	8	10
10	9	9
9	9	10
11	8	11
10	10	9

1. Calculate mean for each worker

$$\mu(\text{Hank}) = 9.6 \text{ min}$$

$$\mu(\text{Joseph}) = 8.8 \text{ min}$$

$$\mu(\text{Susan}) = 9.8 \text{ min}$$

2. Overall mean = 9.4 min.

3. Calculate Sum of square b/w groups (SSB):

$$SSB = n \times (\sum (\bar{x} - x)^2)$$

$$SSB(\text{Hank}) = 5 \times (9.6 - 9.4)^2 = 0.2$$

$$SSB(\text{Joseph}) = 5 \times (8.8 - 9.4)^2 = 1.8$$

$$SSB(\text{Susan}) = 5 \times (9.8 - 9.4)^2 = 0.8$$

$$\text{Total SSB} = 2.8$$

4. Calculate $SSW = \sum (x - \bar{x})^2$

$$\begin{aligned} \text{Hank: } & (8 - 9.6)^2 + (10 - 9.6)^2 + (9 - 9.6)^2 \\ & + (11 - 9.6)^2 + (10 - 9.6)^2 = 5.2 \end{aligned}$$

Similarly

$$\text{Joseph: } 4.8$$

$$\text{Susan: } 1.6$$

$$\text{Total SSW} = 11.6$$

5. Degree of freedom:

$$df \text{ b/w groups} = 2$$

$$df \text{ within groups} = 12$$

6. MSB, Mean Square Between groups

$$MSB = \frac{SSB}{df(\text{between})} = \frac{2.8}{2} = 1.4$$

7. MSW, Mean Square within

$$= \frac{SSW}{df(\text{within})} = \frac{11.6}{12} = 0.9667$$

$$8. F_{\text{stat}} = \frac{MSB}{MSW} = \frac{1.4}{0.966} = 1.44$$

$$9. F_{\text{critical}} \text{ at } 95\% = 3.88$$

$$\therefore F_{\text{stat}} < F_{\text{critical}}$$

Here we do not reject null hypothesis
 $\therefore$ we can say there is not enough evidence to conclude that there are significant differences in the Assembly line task time among numbers.